

One-Step Denoising with MeanFlow Model

基于平均流模型的一步去噪

Abstract

We investigate one-step generative modeling with the recently proposed MeanFlow [1] framework, focusing on its application for image denoising. Our method alters its initial distribution from simple distributions like Gaussian distributions to noisy images, retaining its one-step generation ability while serving the denoising purpose. We adopt the DiT architecture and conduct experiments on MNIST dataset. Results show that a single function evaluation (NFE=1) significantly reduces the noise, raising the Peak Signal-to-Noise Ratio (PSNR) from approximately 6 dB to 22.6 dB. With two evaluations (NFE=2), performance further improves to around 24 dB, whereas additional steps yield little improvement. These findings indicate that the MeanFlow structure is capable of effective denoising at minimal sampling cost, positioning it as a promising practical alternative to conventional multi-step denoising methods.

Methodology

MeanFlow Model [1] defines the average velocity $u(z_t, r, t)$ based on the instantaneous velocity $v(z_t, t)$ in Flow Matching [2]. Their relationship is given by:

$$u(z_t, r, t) \triangleq \frac{1}{t-r} \int_r^t v(z_\tau, \tau) d\tau \quad (1)$$

or equivalently, using the total derivative:

$$u(z_t, r, t) = v(z_t, t) - (t-r) \left[\frac{\partial u}{\partial z_t} v(z_t, t) + \frac{\partial u}{\partial t} \right] \quad (2)$$

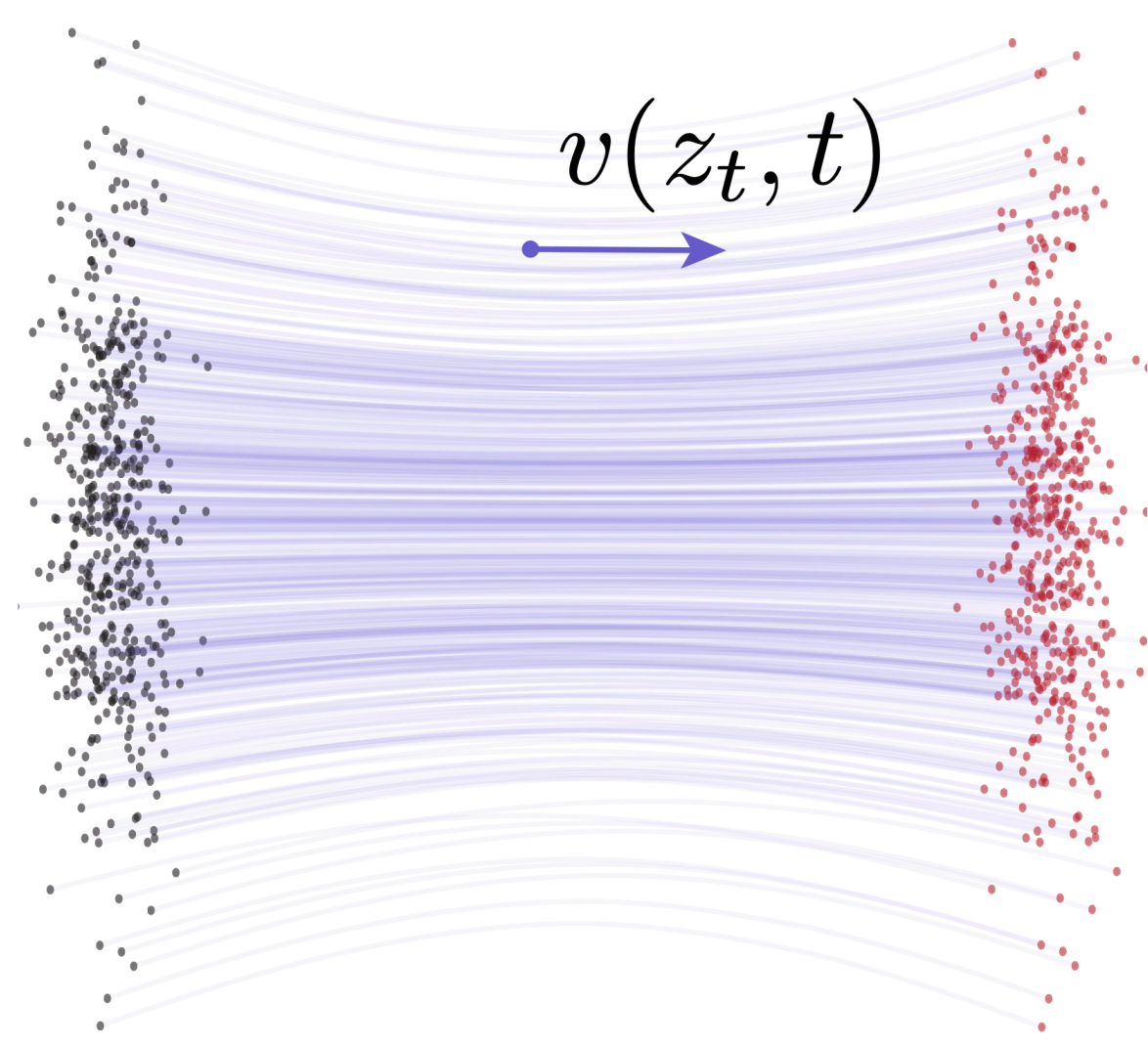


Figure 1: Flow Matching (from [1])

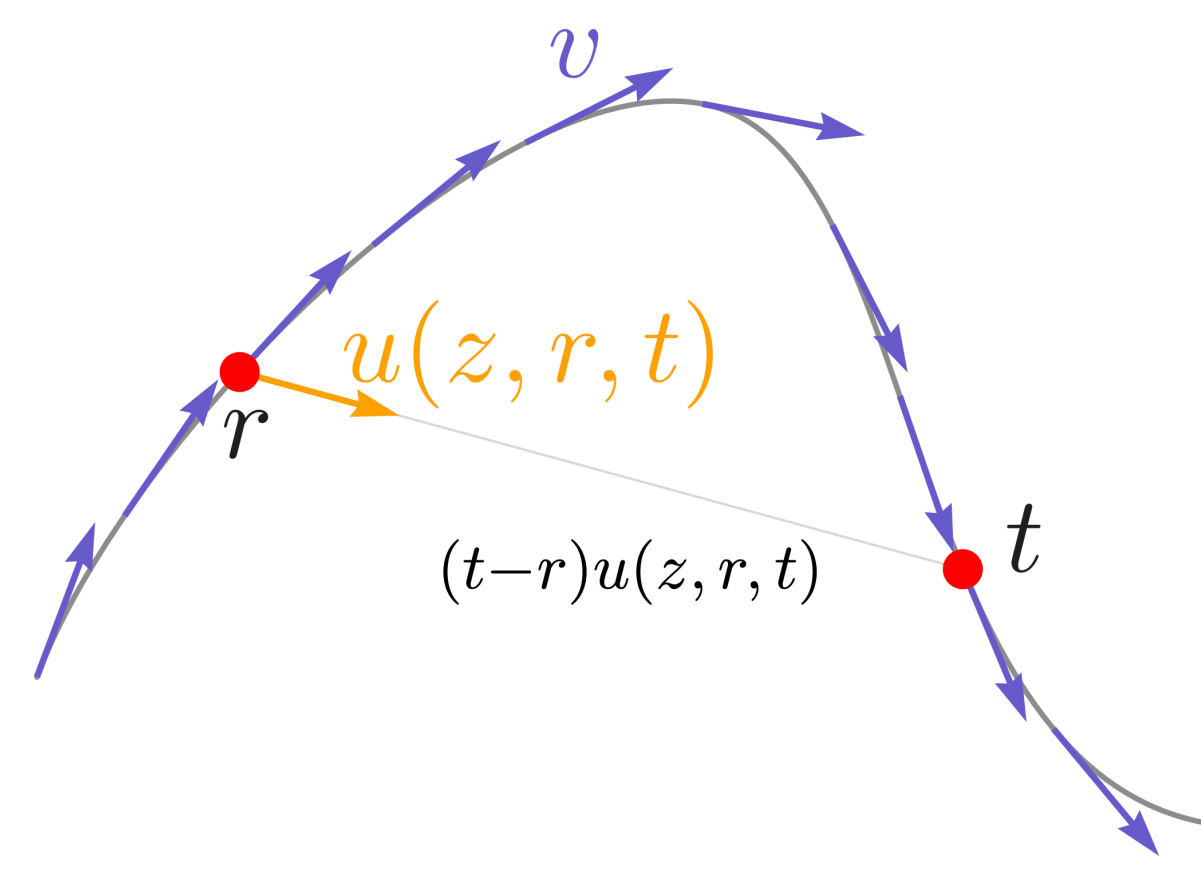


Figure 2: MeanFlow (from [1])

Our model is built within the MeanFlow framework, incorporating classifier-free guidance (CFG) [1] for conditional restoration. The training loss is:

$$\mathcal{L}_{\text{MF}}^{\text{cfg}}(\theta) = \mathbb{E} \left[\left\| u_{\theta}^{\text{cfg}}(z_t, r, t | c) - \text{sg}[u_{\text{tgt}}] \right\|_2^2 \right] \quad (3)$$

where the target u_{tgt} is defined via the CFG version of the conditional instantaneous velocity $\tilde{v}_t$ and the total derivative of the average velocity:

$$u_{\text{tgt}} = \tilde{v}_t - (t-r) \left(\frac{\partial u_{\theta}^{\text{cfg}}}{\partial z_t} \tilde{v}_t + \frac{\partial u_{\theta}^{\text{cfg}}}{\partial t} \right) \quad (4)$$

The total derivative here is computed using **JVP** operation, such as `torch.func.jvp` in PyTorch, and $\tilde{v}_t$ combines the guided and unguided velocities with CFG weight ω :

$$\tilde{v}_t = \omega v_t + (1-\omega) u_{\theta}^{\text{cfg}}(z_t, t, t) \quad (5)$$

Finally, one-step denoising is performed via:

$$z_0 = z_1 - u_{\theta}^{\text{cfg}}(z_1, 0, 1 | c) \quad (6)$$

The training and inference procedures are illustrated in the two Pipelines below, and symbols are summarized in Table 1.

Table 1: Symbols used.

Notation	Meaning
z_t	position at time t
c	CFG labels
z_1	noisy images
x	clean images
t, r	time steps
v_t	conditional instantaneous velocity
ω	guidance scale for CFG
$v(z_t, t)$	instantaneous velocity at (z_t, t)
$\text{sg}[\cdot]$	stop gradient

Figure 3: Pipeline for training.

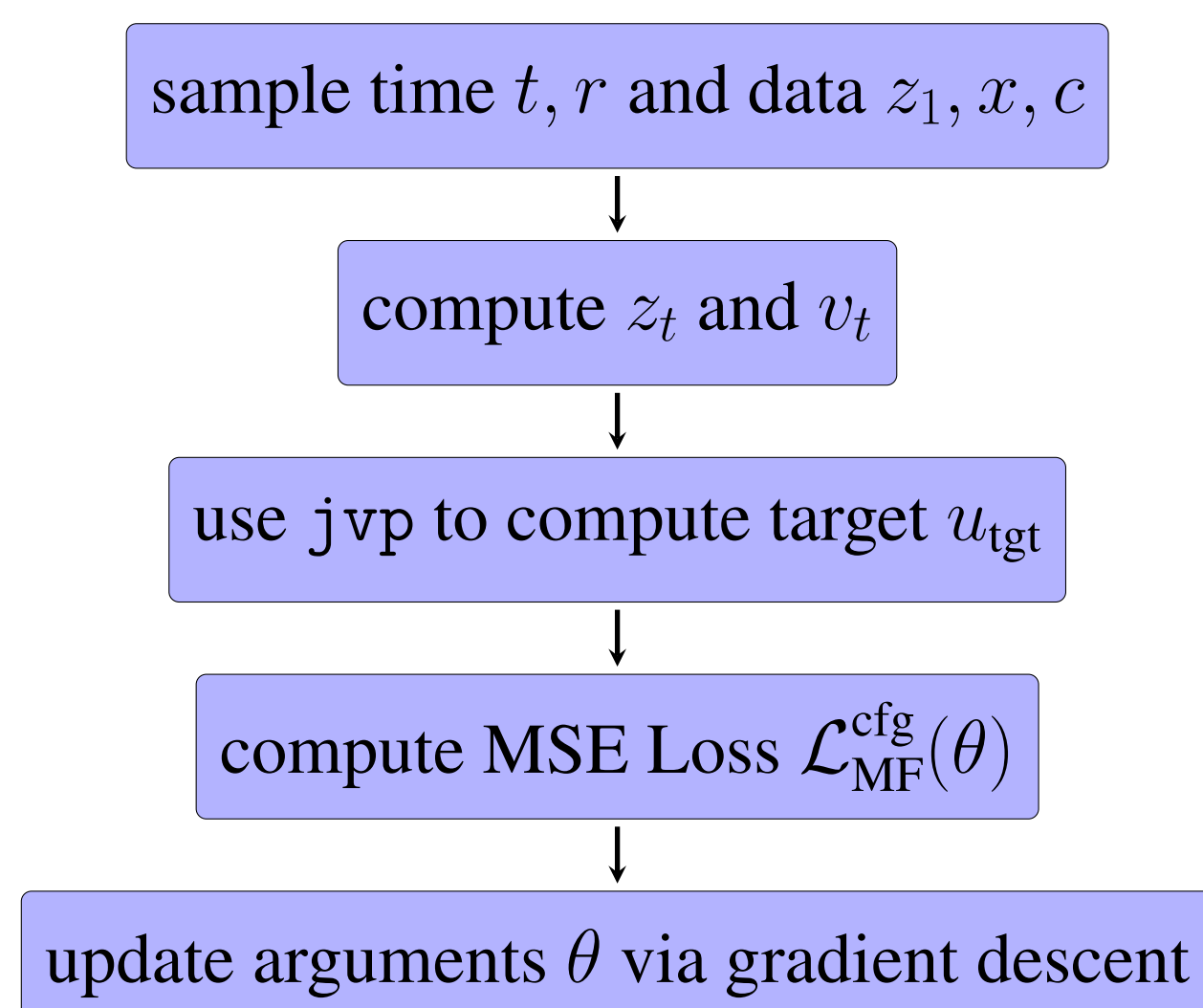
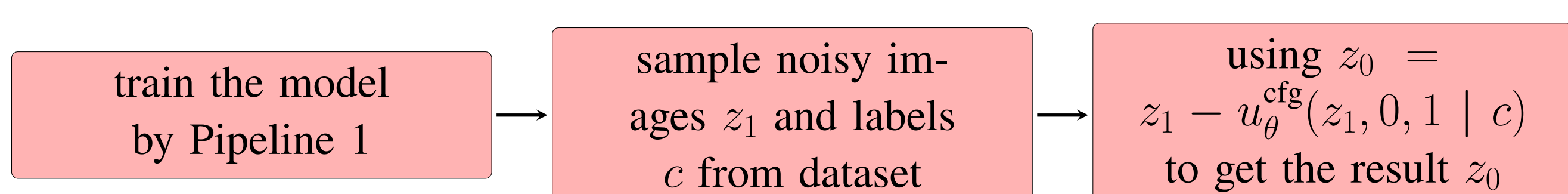


Figure 4: An overall procedure.



Experiment

We implement our image denoising model based on the MeanFlow framework with a DiT architecture and train it using cosine annealing scheduling. As shown in Figure 5, the PSNR values of 1-NFE denoising consistently improve as training progresses. Table 2 summarizes PSNR values at different training stages, while Table 3 compares denoising performance under different NFE after 300k training steps. Visual examples of the denoising results are presented in Figure 6.

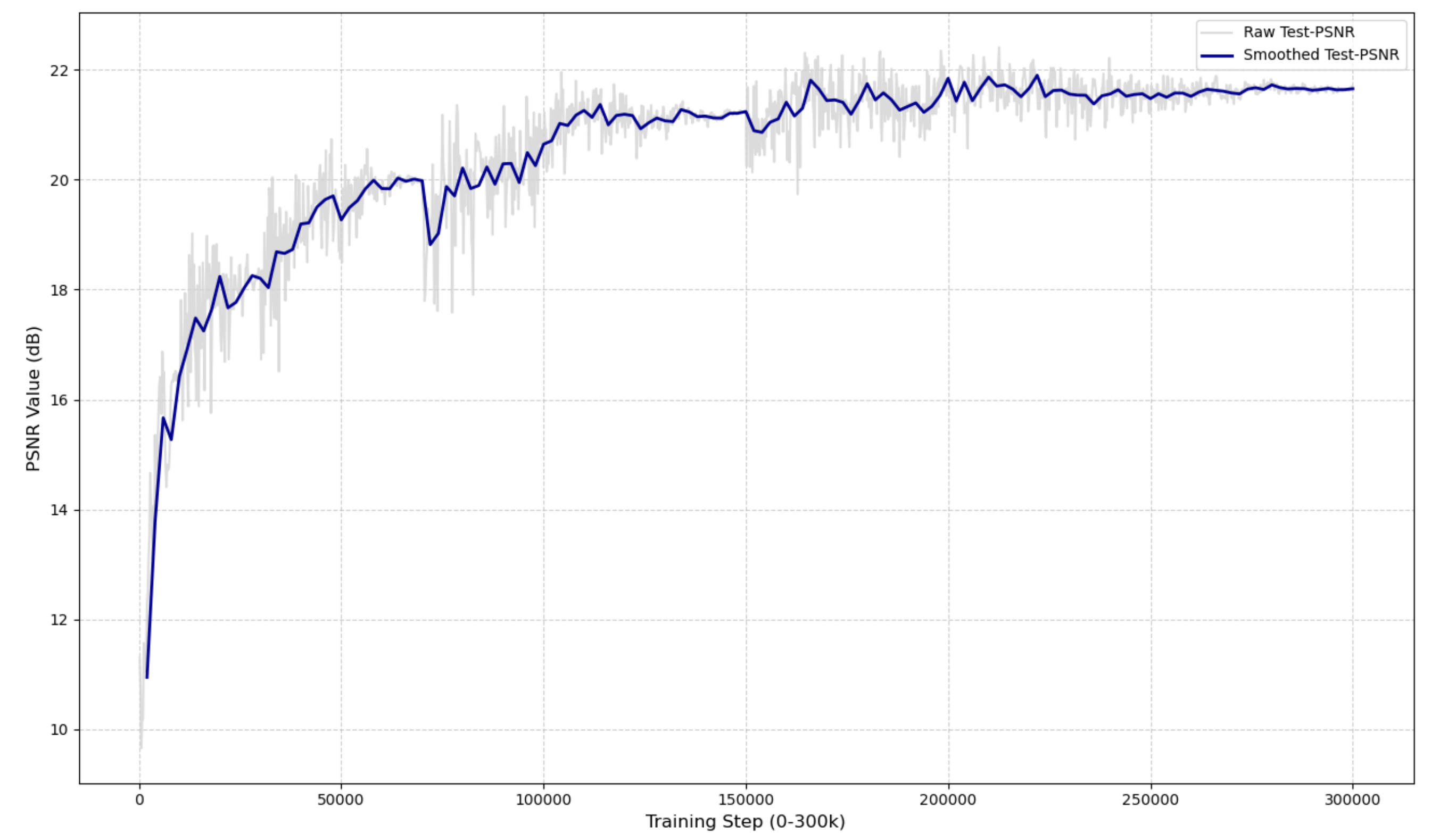


Figure 5: PSNR values over training steps

Images	PSNR (dB)
Noisy Input	5.9312
2k step	12.3043
20k step	17.7944
100k step	20.2236
200k step	21.6694
300k step	21.6807

Table 2: PSNR over training steps

Images	PSNR (dB)
Noisy Input	6.0401
1-NFE	22.6455
2-NFE	24.0686
3-NFE	24.0816
4-NFE	24.0691
5-NFE	24.0558

Table 3: PSNR under different NFE

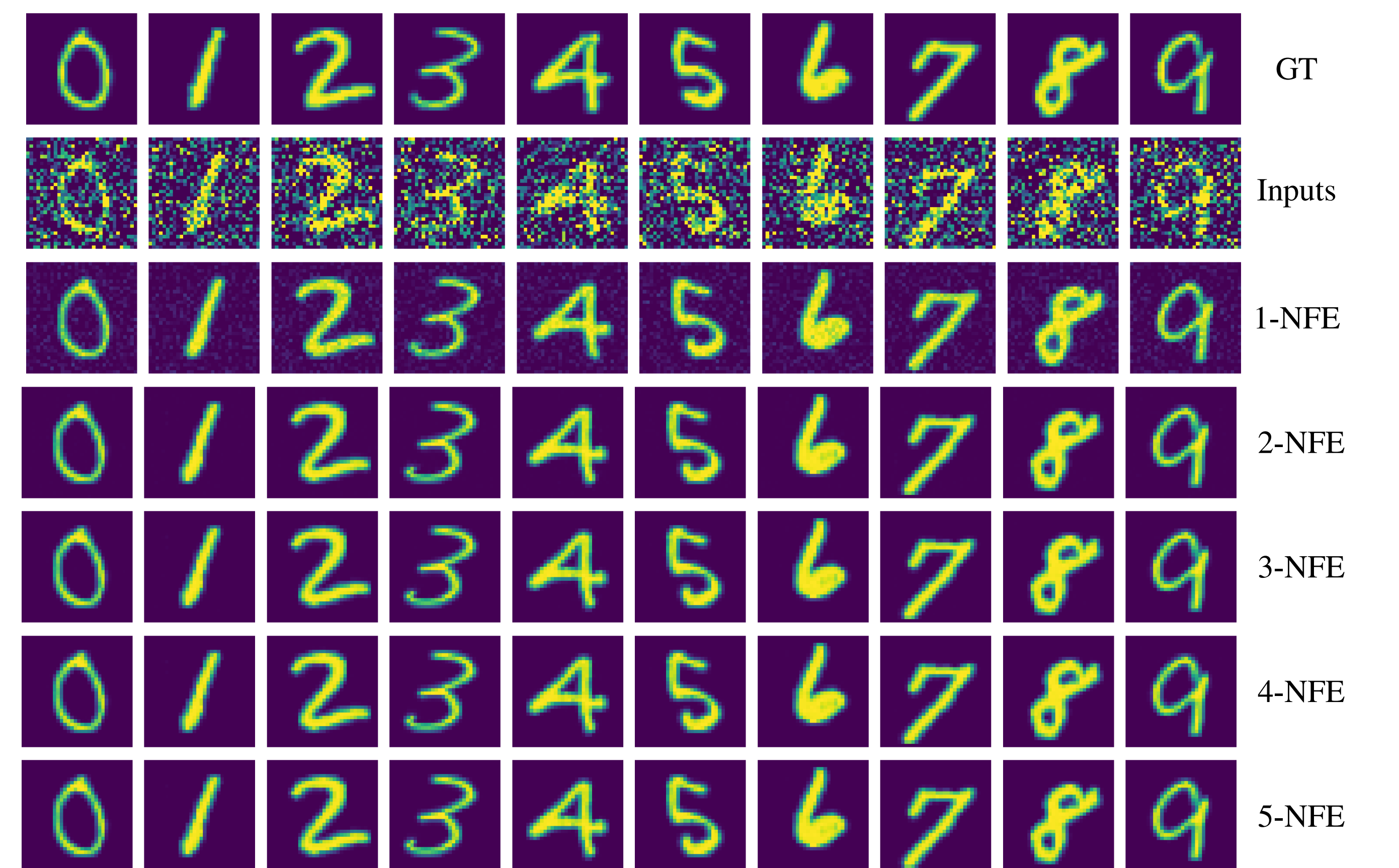


Figure 6: Visual comparison of denoising results under different NFEs. From top to bottom: Ground Truth (clean images), Noisy Inputs, and outputs after 1 to 5 NFE denoising steps.

Conclusion

Our experimental results demonstrate that the MeanFlow framework has considerable potential for image denoising with minimal sampling cost. While the method achieves efficient sampling, the JVP operation incurs significant computational overhead during training. Additionally, the fixed guidance scale in CFG somewhat limits its flexibility. These observations indicate that the MeanFlow architecture would benefit from further improvement, and future work could focus on developing a more computationally efficient and effective training target.

References

- [1] Zhengyang Geng, Mingyang Deng, Xingjian Bai, et al. “Mean Flows for One-step Generative Modeling”. arXiv:2505.13447 [cs.LG], 2025.
- [2] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, et al. “Flow Matching for Generative Modeling”. arXiv:2210.02747 [cs.CV], 2023.